

Phy 403 - Quiz 4 @ 07/11/25

① $1s^2 2p^2 \rightarrow$ last shell $2p^2$ (equivalent e^- 's)
(s-shell filled)

$$L = \sum L_i = |l_1 - l_2| \dots |l_1 + l_2|$$

$$l_1 = l_2 = 1 \Rightarrow l_{\min} = |l_1 - l_2| = 0$$
$$l_{\max} = |l_1 + l_2| = 2$$

$$\therefore L = 0, 1, 2$$

Similarly $S = \sum s_i = |s_1 - s_2|, \dots, |s_1 + s_2|$

$$s_1 = s_2 = \frac{1}{2} \Rightarrow s_{\min} = |s_1 - s_2| = 0$$

$$s_{\max} = |s_1 + s_2| = 1$$

$$\therefore S = 0, 1$$

Pauli exclusion: e^- 's with same $n l m_l m_s$ to be excluded

$$m_{l_1} = m_{l_2} = 1, m_{s_1} = m_{s_2} = \frac{1}{2} \text{ exclude}$$

$\rightarrow e^-$'s are undistinguishable

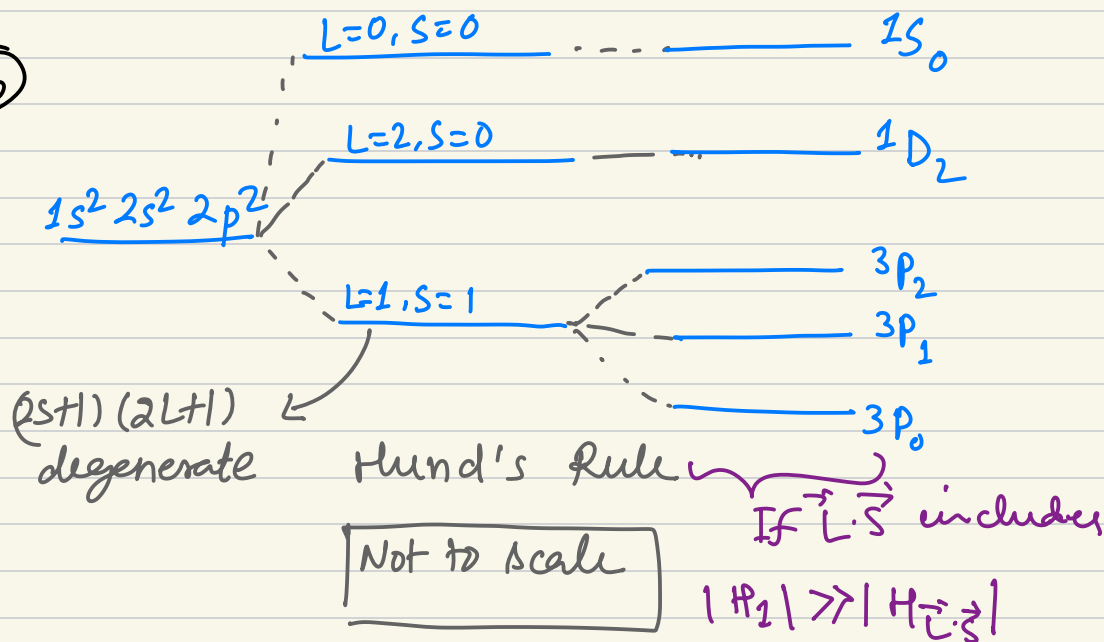
e.g. $m_{l_1} = 1, m_{s_1} = \frac{1}{2}, m_{l_2} = 0, m_{s_2} = \frac{1}{2} \equiv$

$$m_{l_1} = 0, m_{s_1} = -\frac{1}{2}, m_{l_2} = 1, m_{s_2} = \frac{1}{2}$$

$\therefore$ Possible terms are $2S+1 L : 1S \ 3P \ 1D$

(check Quiz-3 1c, Get M_L & M_S & compare)

(1b)



② For a periodic boundary condition

$$e^{i\vec{k} \cdot \vec{r}} = e^{i \frac{2\pi}{L} \hat{n} \cdot \vec{r}} ; \hat{n} \equiv \text{unit vector}$$

$$\therefore k = \frac{2\pi}{L}$$

$$E = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2}{2m} (k_x^2 + k_y^2 + k_z^2)$$

k_x, k_y, k_z defines E space

$$\text{Unit } k \text{ volume} = \left(\frac{2\pi}{L}\right)^3$$

k-space volume corresponding to $E = dk_x dk_y dk_z$
(infinitesimal)

Since k is related to energy ($E \propto k^2$)

$\therefore$ No. of spatial orbital in range $d\vec{k}$

$$= \frac{dk_x dk_y dk_z}{\text{unit } k\text{-volume}} \times 2 \quad \rightarrow \text{spin part of } \sigma$$

$$= \frac{2 k^2 dk d\Omega_k}{(2\pi/L)^3}$$

$$= 2 \cdot \frac{L^3}{8\pi^3} \cdot 4\pi k^2 dk$$

Now, e^- upto $E (= \frac{\hbar^2 k^2}{2m^2})$ will be present

$$\therefore N_s = \text{number of states} = 2 \cdot \frac{L^3}{8\pi^3} \cdot 4\pi \int_0^k k^2 dk$$

$$= \frac{V}{\pi^2} \cdot \frac{1}{3} k^3 = \frac{1}{3\pi^2} \left(\frac{2m}{\hbar^2}\right) V E^{3/2}$$

Now $dN_s = D(E) dE = \frac{dN_s}{dE} dE$

$$\Rightarrow D(E) = \frac{1}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \sqrt{E} V$$

for $T=0$, $E=E_F$ ($\equiv$ Fermi energy) all shells are filled

$$\begin{aligned} \therefore E_{\text{tot}} &= \int_0^{E_F} D(E) E dE = \frac{1}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} V \int_0^{E_F} E^{1/2} E dE \\ &= \frac{1}{5\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} V E_F^{5/2} \end{aligned}$$

Also if N is total no. of electrons, then

$$\begin{aligned} N &= \int_0^{E_F} D(E) dE = \frac{1}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} V \int_0^{E_F} E^{1/2} dE \\ &= \frac{1}{3\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} V E_F^{3/2} \end{aligned}$$

$$\therefore \bar{E} = \frac{E_{\text{tot}}}{N} = \frac{3}{5} E_F$$

③ ② Ground state is stable state of a atom

$$\therefore \tau = \infty$$

① $\tau_a \equiv$ life time of state a

$\tau_b \equiv$ life time of state b

$$\text{As } \Gamma \propto \frac{1}{\tau} \Rightarrow \Gamma = \frac{1}{\tau_a} + \frac{1}{\tau_b}$$

$$\text{or } \Gamma = \hbar \left(\frac{1}{\tau_a} + \frac{1}{\tau_b} \right)$$

$$\Gamma \equiv \text{energy units} \quad e^{\underbrace{Et - \Gamma t/2}}$$

$$\textcircled{c} \quad f(\omega) = \frac{\Gamma^2/4\hbar^2}{(\omega - \omega_0)^2 + \Gamma^2/4\hbar^2}$$

At $\omega = \omega_0$, $f(\omega) = 1$

